

Calculus A(2) Homework 郭嘉乐 2025013332

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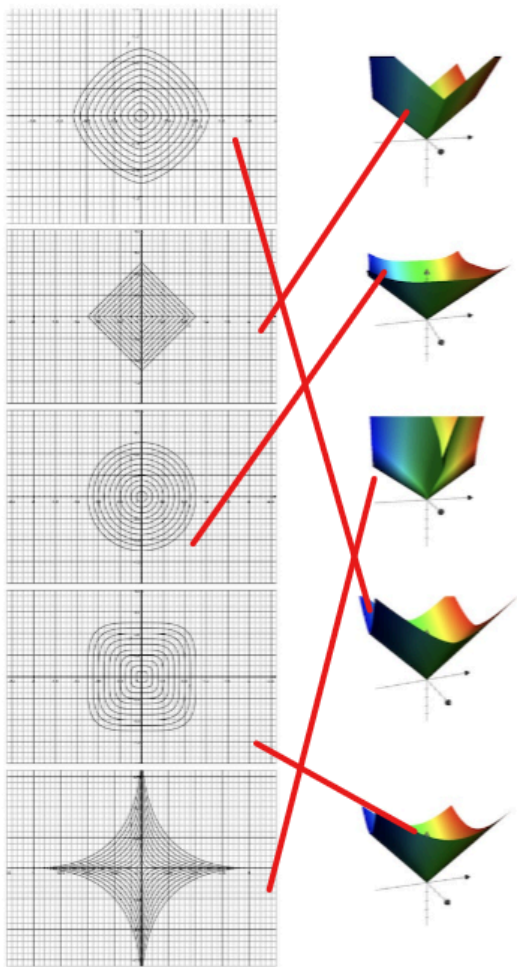
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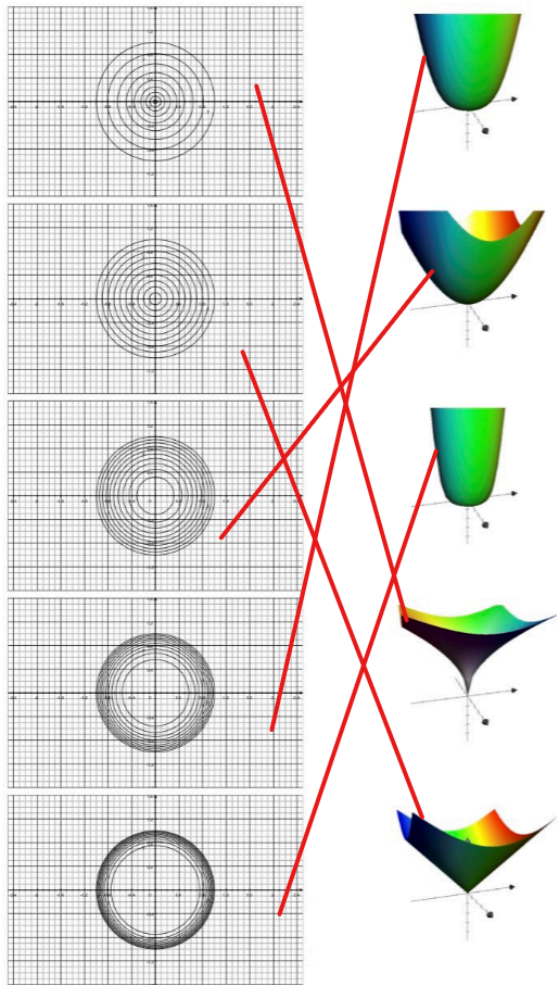
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Problem 2.1



1. For 1st of the left, the level curves are rectangles with rounded corners and their vertices are on the axes, which you can only get it by “cut” the 4th of the right.
2. For 2nd of the left, the level curves are exactly squares with the same center and their vertexs are on the axes, which only fit the 1st of the right.
3. For 3rd of the left, the level curves are circles with the same center, which only fit the 2nd of the right.
4. For 4th of the left, the level curves are rectangles with rounded corners and their sides are parallel to the axes, which you can only get it by “cut” the 5th of the right.
5. For 5th of the left, the level curves like “indents” with their vertexs on the axes, which only fit the 3rd of the right.

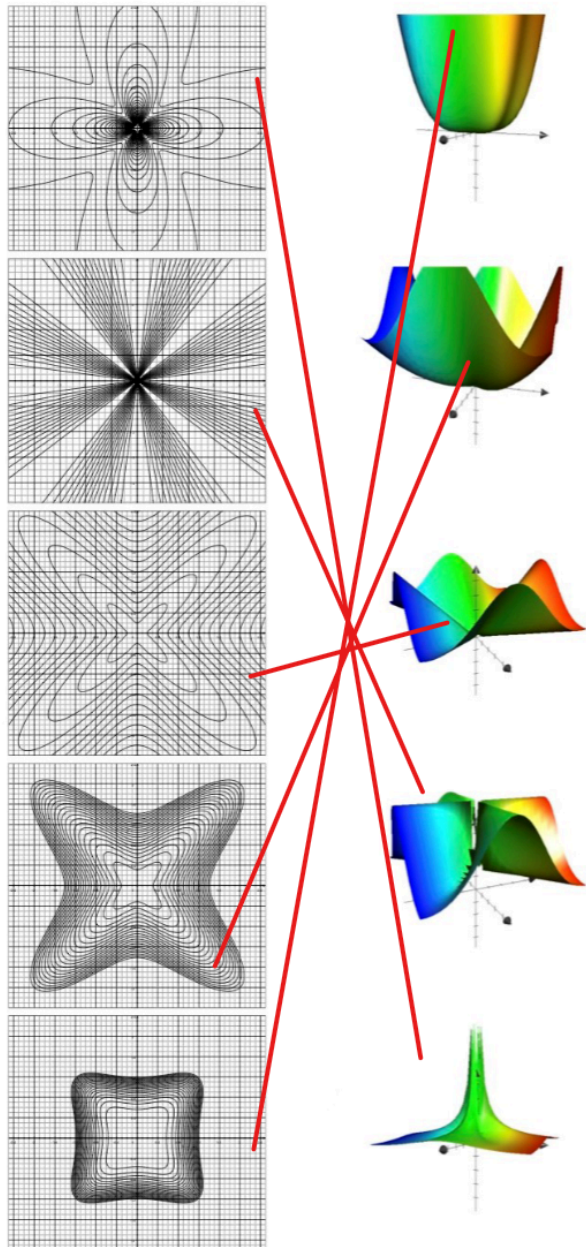
Problem 2.2



(Here the indexes mean the index of the left pictures.)

1. Obviously the density of curves is becoming larger and larger as the distance from the origin increases, which fits 4(R).
2. The density is always the same as the distance from the origin increases, which fits 5(R) as 5(R) looks like a cone.
3. 4. 5. Their densities are becoming larger as the distance from the origin increases. Also, The densities of them are bigger than the one before, and from 3 to 5 and the ratio of the length of the first curve in the middle to the others is also becoming larger, which means 3(L)-2(R), 4(L)-1(R), 5(L)-3(R).

Problem 2.3



(I think it's hard to describe the picture.)

(Here the indexes mean the index of the left pictures.)

1. The only one which the elliptical symmetrical figures of the level curves are on the axes.
- (2.,3.,4.,5. are the same type.)
2. The density is the largest form the origin, and level curves are like radiations.
3. The density is the smallest form the origin.
4. The density is between the 3. and 5.
5. The density is the larger the 4. and the "original" figure is mostly like a square.

Problem 2.4 ****

1. figure 4. see 2.
2. figure 5. for obviously the concave-convex is sharper then 1. as they all have 3 "saddles" and tops.
3. 3. for the last problem we found.
4. figure 2. see in 5.
5. figure 1. for obviously the concave-convex is sharper then 4. as they all have 5 "saddles" and tops.

Problem 2.5

1. We fix z , and get like $y = x^2 - 4z_0 - 2$, it is a parabola. So it is a parabolic cylinder.

2. We get $\frac{x^2}{1} + \frac{y^2}{1} + \frac{z^2}{4} = \frac{1}{4}$. Obviously it is a ellipsoid.

3.* We fix z , then $x^2 + y^2 = \frac{1-z_0}{4}$. It's obviously what we taught: paraboloid.

1. We fix z , then $x^2 - y^2 = \frac{z_0-1}{4}$. From slices to the whole, we find it a hyperbolic paraboloid.

Problem 2.6

1. matches figure 4., $f(r, \theta) = r^3 \cos(3\theta)$, makes the figure smooth and 3 tops and bottoms.

2. matches figure 2., for (let $z = f(x, y)$) $x^2 + y^2 = \ln(\frac{1}{z})$, which compares to 4. the radius of each slice decreases faster as z increases. And they all make bumps. So it is sharper.

3. matches figure 3. for it has $\cos, \sin$ to make the figure have many tops and bottoms by period.

4. matches figure 1., for (let $z = f(x, y)$) $x^2 + y^2 = \frac{1}{z} - 1$. As we say in 2., it is not the shaper one but a bump.

Problem 2.7 *

1. $(x-1)^2 + (y-1)^2 + (z-1)^2 = 4$

2.* Set a point P on the line, a direction vector $\vec{v}$, then it means

$$\begin{aligned} r &= \frac{|\overrightarrow{OP} \times \vec{v}|}{|\vec{v}|} \\ &\Rightarrow 1 = \frac{\left| \begin{pmatrix} x \\ y \\ z \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right|}{\sqrt{1^2 + 1^2 + 1^2}} \\ &\Rightarrow (y-z)^2 + (z-x)^2 + (x-y)^2 = 3 \end{aligned}$$

from the definition.

3. While $x = r \cos \theta$, $y = r \sin \theta$, $r^2 = x^2 + y^2$,

$$\begin{aligned} z &= r^2 \cos(2\theta) \\ &\Rightarrow z = r^2 (\cos^2 \theta - \sin^2 \theta) \\ &\Rightarrow z = (r \cos \theta)^2 - (r \sin \theta)^2 \\ &\Rightarrow z = x^2 - y^2 \end{aligned}$$

4.

$$\begin{aligned} z &= r^3 \cos(3\theta) \\ &\Rightarrow z = r^3 (4 \cos^3 \theta - 3 \cos \theta) \\ &\Rightarrow z = 4(r \cos \theta)^3 - 3r^2(r \cos \theta) \\ &\Rightarrow z = 4x^3 - 3r^2x \\ &\Rightarrow z = 4x^3 - 3(x^2 + y^2)x \\ &\Rightarrow z = x^3 - 3xy^2 \end{aligned}$$

5. $x = \rho \sin \varphi \cos \theta$, $y = \rho \sin \varphi \sin \theta$, $z = \rho \cos \varphi$, $x^2 + y^2 + z^2 = \rho^2$

So

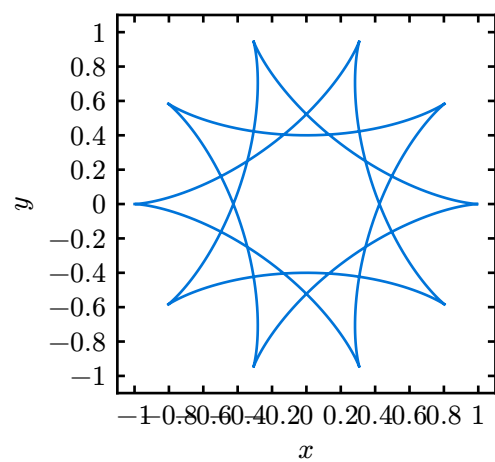
$$\begin{aligned} x &= 2 \cos \theta, y = 2 \sin \theta, z = \rho \cos \varphi \\ &\Rightarrow x^2 + y^2 = 4 \end{aligned}$$

6. Obviously it is $z = 2$.

7.

$$\begin{aligned} \rho \sin^2 \varphi &= \cos \varphi \\ &\Rightarrow \rho \cdot \frac{x^2 + y^2}{\rho^2} = \frac{z}{\rho} \\ &\Rightarrow x^2 + y^2 = z \end{aligned}$$

Problem 2.8 *



No. It's not regular, for from the figure there're many "sudden direction changes", which means there the vector of velocity is 0, so it's not regular.